

# MATT STOKES - Vancouver, BC Canada

Main areas of research and exploration are problem-focused **robotics** and aerospace. [View Portfolio](#)

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## EXPERIENCE

### Nimble Robotics (Full-time) - San Francisco, California

#### Mechanical Engineer, Hardware for New Products Lead Eng Fulltime (Sept 2025 - Present)

- Led architecture, design and implementation of full robotic hardware suite to support VLA/VLM and world model development.
- Types of projects I managed/designed include robotic arms, end effectors, data collection devices, novel one-offs, etc.
  - ◆ Designed device involving entirely novel (patent applied for) mechanical assembly, completely custom EE involving 1x mainboard and 8x sub-boards and readily manufacturable at scale up to 500 units / month.
  - ◆ Designed 6ftx6ftx10ft robotic assembly, for prod deployment, with successful demos run around the USA.

#### Co-op - 4 months (May - Aug. 2025)

- Characterized and redesigned Pneumatics Manifold assembly, assembly time cut 40%, venturi vacuum performance increased 60%.
- Redesigned major robot sub-assembly, performed FEA-based structural analysis, DFA and DFM to drastically cut manufacturing costs while increasing performance.

### Sanctuary AI - Vancouver, BC

#### Mechanical Design Engineer Co-op

##### Co-op - 4 months (Sept. 2024 - Dec. 2024)

- Developed and ran **testing campaign** for a new generation of highly dexterous hands, conducted mechanical **risk analysis**, designed **custom machined electromechanical rigs** to test all main high-risk aspects of new hand design, **summarized** and **modeled** all findings in an **executive facing report**.
- **Designed production components** to address problems identified during my testing campaign, used **GD&T** and **advanced manufacturing** to accomplish ideal cost-to-precision ratios variable on component/assembly requirements.
- Developed **spreadsheet-based mathematical models** for **force output**, **wear**, and **backlash** prediction over time for all major mechanical sub-assemblies of the new generation of hand.

### Rugged Robotics - Houston, TX

#### Mechanical Engineering Co-op

##### Co-op - 4 months (Jan. 2024 - April. 2024)

- Redesigned a full **subsystem** of the Mk1 robot platform, incorporating injection molded components to reduce **cost** of the sub-system by **~83%**.
- Designed and implemented high-precision laser alignment and characterization rig, utilizing **Solidworks**, **SW FEA** and **GD&T**. Reduced localization error rooting from measurement sensor modules by several orders of magnitude.

**PREVIOUS PROJECTS -- [See my portfolio!](#)**

## EDUCATION

### Honors Bachelors of Mechatronics Engineering

University of Waterloo, ON

2023-2029

#### Additional Previous Positions/Experience

- **Prosper Robotics**
  - ◆ Remote Contractor
  - ◆ 3-mo
  - ◆ Designed Parallel End Effector, 50N force output, Series Elastic Compliance for human interfacing
- **GoogleX (X Moonshot Factory)**
  - ◆ Student Researcher (Highschool)
  - ◆ 1mo
  - ◆ Worked with Rapid Eval and Design Kitchen Team on a Moonshot Concept

## CURRENT PROJECTS

- **Humanoids**
  - ◆ Currently on V4 of a humanoid platform I have been working on for ~3 years
  - ◆ V1-V2 designed in Solidworks, V3 onwards based in Onshape.
- **BIG Drone**
  - ◆ Custom composite frame heavy-lift drone (180lb payload)
- **UWATERLOO HACKERFAB**
  - ◆ Building a semiconductor fab facility at Waterloo University.
  - ◆ Lithography and photoresist based, single layer integrated circuit fabrication.